Technext

PROTOTYPE

CLIENT GUIDE 01

The database, explained

Follow one number from a company to a report card

Prepared by Technext for EnergieBee · 26 October 2026 · examples come from the real UK home

THIS IS A PROTOTYPE. A working model built to prove the ideas, not a finished product.

- It was built and tested for **one real home**. There is no sign-in.
- **France and the Netherlands**: the database has settings for them (money, time zone, usual box size), but no real data, no tax rules and no prices yet.
- The helpers that decide *when to charge a battery or heat a house* are **not built**. The database already has room for them.

The idea

The database is a **diary of the home's energy**. It follows three rules: **keep the original** (like a shop receipt), **write down who told us and when**, and **never erase**. If a number is corrected, the new one is added and the old one is kept, marked "replaced".

The story in five steps

1. **A company sends us numbers.** For example, at 13:00 on 19 October the solar box said "3,713 watts".

Who sends	What they send	Status
Sunsynk	The solar and battery box: power every 5 minutes, and daily totals	**Connected**
Octopus	The meters, the home's tariff, prices and smart-charge times	**Connected**
Tariffscanner	Other UK companies' tariffs. Stored, not yet used to price the home	**Connected**
NESO, National Gas, Open-Meteo, Quartz	How clean the grid is, the energy in gas, the weather, the sun forecast	**Connected**
tounify	Prices for France and the Netherlands	**Built, off until its licence is confirmed**
ENTSO-E, Linky, P1	EU market prices and the French and Dutch smart meters	**Planned, not built**

"Connected" means tested end to end on the one real UK home on 26 October 2026.

2. **We keep the original.** The answer is saved as a file, exactly as it arrived. One row in the table `raw_payload` is its **receipt**: who sent it, when we fetched it, the file name, and whether we accepted it.
3. **We write it in the diary.** A **reading** is a value at one moment. An **interval** is an amount over a period, like a meter's half-hour. A **forecast** is a prediction. Each one carries its receipt number. Readings and intervals also say when we heard them and when they were replaced.
4. **We cut time into boxes, called slots.** Each home uses 15, 30 or 60 minute boxes. For every box we add up the energy, work out who fed whom, and price it. Page 3 shows how, with a real example.
5. **We check it and add a label.** A box is **final** (everything needed is in), **waiting** (the meter has not reported yet, so the solar box fills in) or **missing** (no data; we never invent it). The finished boxes are rows in the table `site_hh`.

The map of the database

This map shows where each step lives. Each box is a **table**, like a page of a spreadsheet. A line means one table points at another. The **orange dashed arrows** are not stored links: they show the report cards being *computed* from the diary (prices and tariffs feed them too, but are not drawn). Read it from left to right: the companies' answers, the receipts, the diary, the report cards.



Figure 1. Generated from the real schema as it stood on 26 October 2026: all 12 solid arrows are real links. Black headers are tables written in `src/db/schema.sql`; orange ones were added later by delta files. PK marks the columns that identify a row.

The whole database has 54 tables: 23 in `schema.sql` and 31 added by delta files. Page 4 lists them all.

Step 4 in detail: how 15, 30 and 60 minute boxes are made

A box is one row of the table `site_hh`. ("hh" is short for half-hour, the first box size. The table now holds 15 and 60 minute boxes too.) A row has 52 columns in seven groups:

Group	What it holds	Columns
Which box	which home, when the box starts and how long it lasts	5
Can I trust it	final, waiting or missing, plus notes on how it was made	7
Energy measured (kWh)	how much solar, grid, battery and house use there was	10
Who fed whom (kWh)	for example how much solar went to the house	10
Prices	the price per kWh used for this box	5
Costs and savings	what the box cost, and what was saved	10
Carbon and weather	how clean the power was, the weather, the sun forecast	5

A real example: the real UK home, 24 October 2026, 05:00 to 06:00

Two sources describe this hour. The **solar box** measures grid power every 5 minutes (the orange bars, in watts). The **smart meter** gives the official figure for each half-hour. The boxes are built from those:

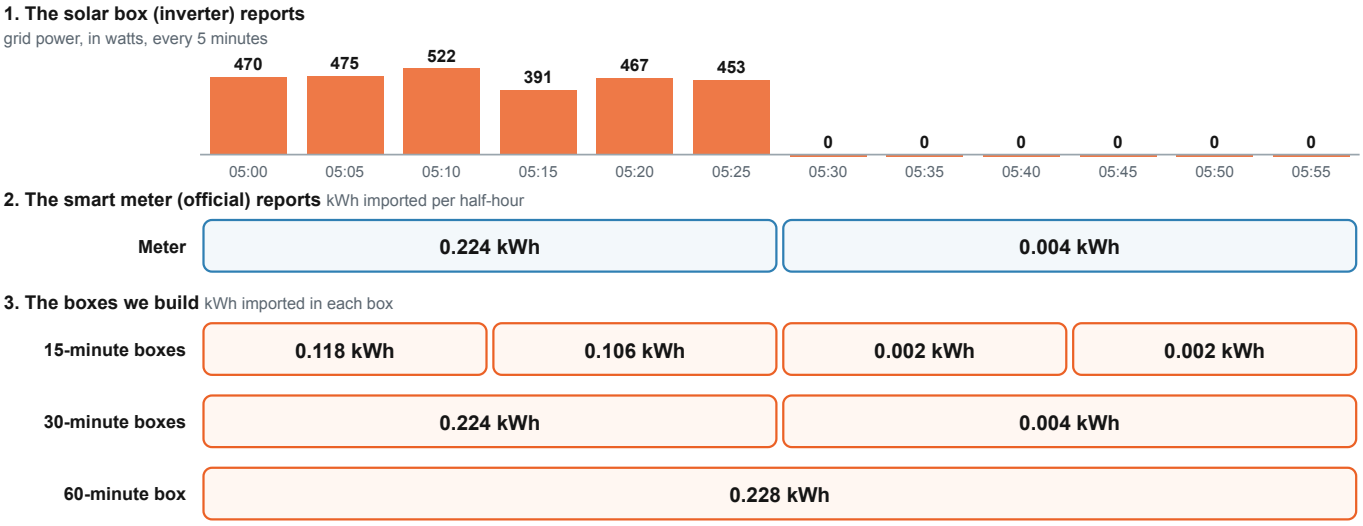


Figure 2. Real data, one hour. Every row adds up to 0.228 kWh.

1. **30-minute boxes** are the meter's figures as they are: 0.224 and 0.004 kWh.
2. **A 60-minute box** adds the two: 0.224 + 0.004 = 0.228 kWh.
3. **15-minute boxes** split each meter half-hour in two, in the same proportion as the solar box's readings.
   05:00 to 05:30: the solar box's readings add up to 1,467 for the first quarter-hour and 1,311 for the second (470 + 475 + 522 and 391 + 467 + 453). So the meter's 0.224 kWh is split 53% and 47%: **0.118** and **0.106** kWh.
   05:30 to 06:00: the solar box saw no power, so the meter's 0.004 kWh is split equally: **0.002** and **0.002** kWh.

Prices. This home pays **7.619 p** per kWh until 05:30 and **32.144 p** after (before VAT). A 15 or 30 minute box sits on one side of 05:30, so it has one clear price. A 60-minute box covers both, so it uses the **average**: 19.882 p.

> **Careful.** Almost all the power was used in the cheap half-hour. Priced in 30-minute boxes, the hour costs **1.84 p**. Priced as one 60-minute box, it costs **4.53 p**, which is 2.70 p too much, because the big box cannot see when the power was used. So a box should not be longer than the price steps. This home uses 30-minute boxes.

On the days the clocks change, a day has 23 or 25 hours. With 30-minute boxes that is 46 or 50 boxes instead of 48 (15-minute boxes: 92 or 100 instead of 96; 60-minute: 23 or 25 instead of 24). If a home changes its box size, the boxes are simply rebuilt from the diary and nothing is lost.

The whole database on one page

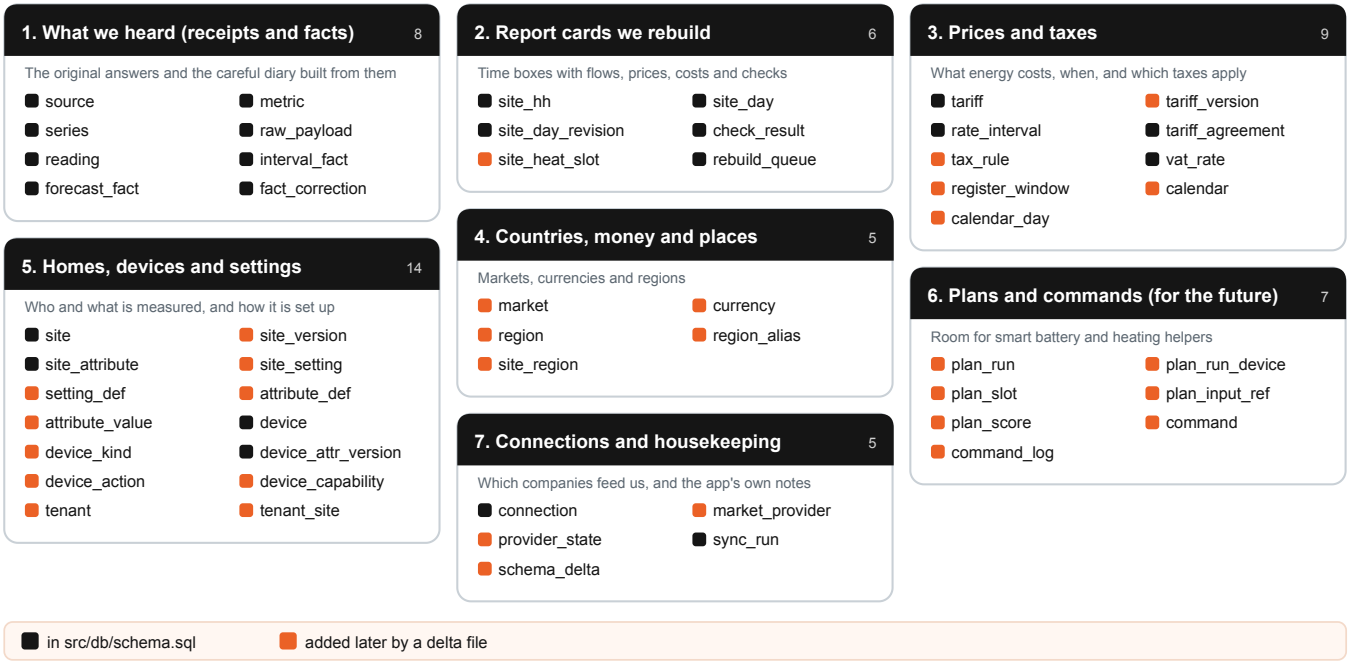


Figure 3. All 54 tables, grouped by job. Black: in `schema.sql`. Orange: added by a delta file.

Can I trust it?

- **Receipts.** Facts carry a receipt number that points at the saved file. (A rule can tidy up old unused files, but it only reports unless someone switches it on.)
- **Labels.** Every price, tax and rate says where it came from: *sourced*, *from the provider*, *typed in*, *placeholder* or *practice data*.
- **Double-checks.** The meter is compared with the solar box, energy in with energy out, and missing prices are looked for. Failures are shown, not hidden.
- **A time machine.** Any day can be rebuilt as we knew it on an earlier date.

What works, and what does not

Works today	The diary, the receipts and the boxes for the real UK home, with checks, labels and the time machine.
Set up, but no real data	France and the Netherlands: money in cents, time zone and usual box size are set up (marked "placeholder"). There are no tax rules, prices or meters for them yet; the tests use made-up homes.
Not built	The battery and heating helpers, sending commands to devices, sign-in, and many homes at once. The database already has a place for all the data the helpers need (checked item by item: 456 of 456 verified items covered, 0 gaps).